

## ABU SIMBEL, Egypt

WMO: 624190

Lat: 22.376N

Lon: 31.612E

Elev: 616

StdP: 14.37

Time Zone: 2.00 (E02)

Period: 06-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | 46.5       | 48.5 | 17.2                          | 13.4 | 70.6 | 20.8 | 16.0 | 68.0 | 21.3                  | 64.5 | 19.4 | 65.0 | 6.0                   | 340  | 0.478 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |       |      |       |      | Evaporation WB/MCDB |       |      |       |      |       | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|-------|------|-------|------|---------------------|-------|------|-------|------|-------|----------------------|------|
|               |                        | 0.4%            |      | 1%    |      | 2%    |      | 0.4%                |       | 1%   |       | 2%   |       |                      |      |
|               |                        | DB              | MCWB | DB    | MCWB | DB    | MCWB | WB                  | MCDB  | WB   | MCDB  | WB   | MCDB  | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)   | (f)  | (g)   | (h)  | (i)                 | (j)   | (k)  | (l)   | (m)  | (n)   | (o)                  | (p)  |
| 8             | 25.6                   | 111.2           | 70.6 | 109.4 | 70.0 | 107.6 | 69.5 | 74.5                | 100.7 | 72.9 | 101.2 | 71.8 | 102.2 | 10.0                 | 60   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |       |      |       |      |       | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|-------|------|-------|------|-------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |       | 1%   |       | 2%   |       |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB  | Enth | MCDB  | Enth | MCDB  |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)   | (l)  | (m)   | (n)  | (o)   | (p)               |
| 65.2                            | 95.3 | 93.4 | 62.2 | 85.6 | 93.0 | 59.3 | 77.0 | 91.0 | 38.3          | 100.6 | 36.9 | 102.0 | 35.8 | 101.8 | 79.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |     | n-Year Return Period Values of Extreme Temperature |     |            |     |            |     |            |     |  |  |
|-------------------|------|------|----------------------------|-----|--------------------|-----|----------------------------------------------------|-----|------------|-----|------------|-----|------------|-----|--|--|
|                   |      |      | Mean                       |     | Standard Deviation |     | n=5 years                                          |     | n=10 years |     | n=20 years |     | n=50 years |     |  |  |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max | Min                                                | Max | Min        | Max | Min        | Max | Min        | Max |  |  |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h) | (i)                                                | (j) | (k)        | (l) | (m)        | (n) | (o)        | (p) |  |  |
| (4) 19.7          | 17.8 | 16.1 | N/A                        | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
| (5)               |      |      | DB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |
|                   |      |      | WB                         | N/A | N/A                | N/A | N/A                                                | N/A | N/A        | N/A | N/A        | N/A | N/A        | N/A |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 80.6   | 62.0  | 66.7  | 73.3  | 81.6  | 88.5  | 93.2  | 92.8  | 94.6  | 90.6  | 84.7  | 73.8  | 64.4  |
|                                                                        | DBStd       | 12.60  | 5.57  | 6.17  | 6.92  | 6.21  | 5.00  | 3.77  | 3.19  | 2.97  | 3.64  | 5.97  | 5.50  | 4.87  |
|                                                                        | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     |
|                                                                        | HDD65       | 257    | 131   | 45    | 8     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 2     | 71    |
|                                                                        | CDD50       | 11162  | 373   | 467   | 722   | 948   | 1194  | 1295  | 1328  | 1383  | 1218  | 1076  | 713   | 445   |
|                                                                        | CDD65       | 5944   | 38    | 92    | 265   | 498   | 729   | 845   | 863   | 918   | 768   | 611   | 266   | 51    |
|                                                                        | CDH74       | 92060  | 419   | 1039  | 3206  | 6827  | 11512 | 14421 | 14841 | 15887 | 12261 | 8433  | 2755  | 459   |
|                                                                        | CDH80       | 60178  | 114   | 375   | 1515  | 3956  | 7616  | 10236 | 10511 | 11483 | 8193  | 5006  | 1091  | 82    |
| (14) Wind                                                              | WSAvg       | 8.0    | 7.6   | 8.0   | 8.3   | 7.9   | 8.3   | 8.2   | 6.8   | 7.8   | 8.7   | 8.3   | 7.8   | 8.1   |
| (15) Precipitation                                                     | PrecAvg     | 0.0    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecMax     | 0.2    | 0.2   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecMin     | 0.0    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
|                                                                        | PrecStd     | 0.0    | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   | 0.0   |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 89.3  | 95.7  | 103.5 | 109.2 | 111.3 | 114.5 | 112.1 | 112.5 | 109.8 | 107.8 | 96.5  | 86.0  |
|                                                                        |             | MCWB   | 64.6  | 64.9  | 67.8  | 68.6  | 69.1  | 70.3  | 70.6  | 71.0  | 69.9  | 69.6  | 68.4  | 63.4  |
|                                                                        | 2%          | DB     | 82.7  | 89.3  | 98.3  | 104.2 | 108.7 | 111.2 | 109.5 | 110.2 | 107.2 | 104.0 | 91.8  | 82.3  |
|                                                                        |             | MCWB   | 60.5  | 61.9  | 65.3  | 66.6  | 68.7  | 70.1  | 69.8  | 71.4  | 69.8  | 69.4  | 66.7  | 61.1  |
|                                                                        | 5%          | DB     | 78.7  | 84.4  | 93.1  | 100.0 | 105.8 | 108.0 | 107.6 | 108.1 | 104.4 | 100.7 | 89.3  | 79.0  |
|                                                                        |             | MCWB   | 58.0  | 59.4  | 63.0  | 65.2  | 67.8  | 69.1  | 69.4  | 70.7  | 68.9  | 68.7  | 65.7  | 59.6  |
|                                                                        | 10%         | DB     | 74.8  | 80.5  | 87.9  | 96.6  | 102.5 | 106.0 | 105.7 | 106.1 | 102.5 | 97.6  | 86.2  | 75.7  |
|                                                                        |             | MCWB   | 56.2  | 58.0  | 60.1  | 63.6  | 66.3  | 68.1  | 68.8  | 70.1  | 68.6  | 68.3  | 63.8  | 57.9  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 65.9  | 67.0  | 69.7  | 70.2  | 73.3  | 73.9  | 76.5  | 76.9  | 73.7  | 73.7  | 71.2  | 65.7  |
|                                                                        |             | MCDB   | 85.1  | 90.1  | 96.7  | 102.6 | 102.8 | 104.1 | 101.9 | 101.9 | 102.0 | 98.3  | 90.2  | 80.5  |
|                                                                        | 2%          | WB     | 62.0  | 62.9  | 66.4  | 68.0  | 70.9  | 71.9  | 73.2  | 75.0  | 72.1  | 71.7  | 69.2  | 63.1  |
|                                                                        |             | MCDB   | 80.7  | 86.1  | 94.3  | 98.4  | 102.9 | 105.5 | 102.0 | 100.5 | 101.9 | 97.8  | 89.6  | 79.2  |
|                                                                        | 5%          | WB     | 59.0  | 60.4  | 63.8  | 66.2  | 69.0  | 70.5  | 71.1  | 73.4  | 70.7  | 70.4  | 66.9  | 61.0  |
|                                                                        |             | MCDB   | 77.2  | 82.8  | 90.5  | 96.8  | 101.7 | 105.1 | 103.5 | 99.9  | 101.2 | 96.8  | 86.6  | 76.9  |
|                                                                        | 10%         | WB     | 56.4  | 58.3  | 61.3  | 64.5  | 67.4  | 69.2  | 69.8  | 72.1  | 69.6  | 69.0  | 64.8  | 58.7  |
|                                                                        |             | MCDB   | 72.8  | 79.2  | 87.0  | 94.1  | 100.1 | 103.9 | 103.0 | 101.9 | 100.1 | 95.5  | 84.3  | 74.8  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 23.8  | 24.7  | 26.9  | 26.6  | 26.8  | 26.5  | 26.5  | 25.6  | 25.2  | 23.9  | 23.9  | 23.2  |
|                                                                        |             | MCDBR  | 27.6  | 29.4  | 31.0  | 30.2  | 30.1  | 27.8  | 27.9  | 26.4  | 26.3  | 26.0  | 24.6  | 24.7  |
|                                                                        |             | MCWBR  | 14.6  | 14.2  | 13.3  | 12.0  | 12.4  | 11.1  | 10.9  | 10.3  | 10.5  | 11.0  | 12.7  | 13.1  |
|                                                                        | 5% WB       | MCDBR  | 27.2  | 28.7  | 29.9  | 29.2  | 27.8  | 26.0  | 25.9  | 23.6  | 24.9  | 24.6  | 23.5  | 23.2  |
|                                                                        |             | MCWBR  | 14.7  | 14.4  | 13.3  | 12.4  | 12.2  | 11.1  | 11.5  | 10.6  | 11.2  | 11.3  | 13.2  | 12.9  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.379 | 0.422 | 0.464 | 0.553 | 0.575 | 0.527 | 0.548 | 0.578 | 0.540 | 0.499 | 0.407 | 0.383 |
|                                                                        | taud        |        | 2.152 | 2.006 | 1.880 | 1.694 | 1.666 | 1.790 | 1.773 | 1.729 | 1.792 | 1.871 | 2.083 | 2.134 |
|                                                                        | Ebn at Noon |        | 278   | 273   | 266   | 245   | 238   | 248   | 243   | 236   | 242   | 246   | 266   | 272   |
|                                                                        | Edh at Noon |        | 44    | 54    | 64    | 78    | 80    | 70    | 71    | 75    | 69    | 61    | 46    | 43    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1610  | 1882  | 2210  | 2433  | 2509  | 2558  | 2485  | 2344  | 2178  | 1917  | 1666  | 1521  |
|                                                                        | RadStd      |        | 43    | 54    | 54    | 52    | 54    | 34    | 40    | 43    | 43    | 26    | 29    | 23    |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (0 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page